A Rigorous Proof of Fermat's Theorem: Complete Coverage of Natural Numbers and the Impossibility of Integer Solutions for n>3n > 3n>3
The Structure of Fibonacci Numbers and the Principles of the Golden Ratio as the Foundation of the Final Proof

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Annotation:
The model of the universe is considered as a simply connected three-dimensional fractal manifold, where the key parameters—speed of light c=300,000 km/sc = 300,000 \, \text{km/s}c=300,000km/s, time, space, gravity, and energy—are interconnected within a unified system. This interaction creates a dynamic harmony, explained through the following principles.

$$\Psi universe(t)$$

$$\Psi universe(t) = \sum n=1NFn / (4/3\pi(R0e^Ht)^3) + E/c^2 \cdot 1/\rho(t)$$

Where:
1. Ψuniverse(t) — the universal wave function describing all of reality at time t.
2. ∑n=1NFn / (4/3π(R0e^Ht)^3) — the energy density distributed within the expanding space according to the Fibonacci sequence:
   - R_0 — the initial radius ($R_0 \to 0$).
   - H — the Hubble constant.
   - t — time (absolute or embedded).
   - F_n — Fibonacci numbers reflecting the fractal nature of the distribution.
3. E/c^2 — the equivalent mass of the entire system via the energy-mass relation (as per Einstein).
4. 1/ρ(t) — the inverse density, representing the interaction of matter and void.

This formula describes the expansion of the light sphere, which maintains an even distribution of energy and is confirmed by data on the observed anisotropy of the cosmic microwave background (CMB). Empirical data from the Planck, COBE, and WMAP missions have shown consistency between the fractal universe model and observations of small and large temperature fluctuations in the CMB. These fluctuations are interpreted as a result of internal energy requantization. The distribution of energy within the universe adheres to the Fibonacci harmony. Empirical evidence supports this through data on galactic structure and dark matter density obtained via gravitational lensing maps from the Hubble, Gaia, and JWST missions. The speed of light **ccc** is explained as a parameter that maintains the balance between the expansion of space and the internal dynamics of energy. This is confirmed by observations of the coherent expansion of the universe and the distribution of energy on large scales. Power spectrum data from the CMB indicate the preservation of proportions in large-scale structures."

Conclusion:
Thus, the proof of Fermat's theorem is based on the simply connected fractal structure of the universe, where numbers and forms are part of the harmony of space-time dynamics. The model is validated both mathematically and empirically through observations of the cosmic microwave background, energy distribution, and galactic structures.

1. Problem Statement
Fermat's Last Theorem states that for all n>2n > 2n>2, the equation
xn+yn=znx^n + y^n = z^nxn+yn=zn
has no integer solutions in positive natural numbers x,y,zx, y, zx,y,z.

2. Coverage of All Natural Numbers
🔷 According to **Zechendorf's Theorem**, any natural number can be represented as the sum of non-overlapping Fibonacci numbers:
n=Fk1+Fk2+⋯+Fkm,k1>k2>⋯>km.n = F_{k_1} + F_{k_2} + \dots + F_{k_m}, \quad k_1 > k_2 > \dots > k_m.n=Fk1+Fk2+⋯+Fkm,k1>k2>⋯>km.
For example:
27=21+5+1,27 = 21 + 5 + 1,27=21+5+1,
where 21,5,121, 5, 121,5,1 are Fibonacci numbers.
Thus, the Fibonacci sequence fully covers all natural numbers.

3. Irrationality in the Structure of Fibonacci Numbers
🔷 Fibonacci numbers can be expressed through Binet's formula:
Fn=φn−(1−φ)n5,F_n = \frac{\varphi^n - (1 - \varphi)^n}{\sqrt{5}},Fn=5φn−(1−φ)n,
where φ≈1.618\varphi \approx 1.618φ≈1.618 is the golden ratio, and 1−φ≈−0.6181 - \varphi \approx -0.6181−φ≈−0.618.
When raised to a power n>3n > 3n>3, the formula becomes:
Fnn=(φn−(1−φ)n5)n.F_n^n = \left(\frac{\varphi^n - (1 - \varphi)^n}{\sqrt{5}} \right)^n.Fnn=(5φn−(1−φ)n)n.
An irrational component (1−φ)n(1 - \varphi)^n(1−φ)n appears and persists under any transformations, making it impossible to express the result as a rational number.

4. Contradiction with Integer Solutions
🔷 Suppose there exist positive integers x,y,zx, y, zx,y,z that satisfy Fermat's equation for n>3n > 3n>3: xn+yn=zn.x^n + y^n = z^n.xn+yn=zn.

Since x,y,zx, y, zx,y,z can be expressed as sums of Fibonacci numbers, their powers will include irrational residues based on Binet's formula.
🔷 However, the equation requires that the sum of xn+ynx^n + y^nxn+yn and znz^nzn must be a whole number. This is impossible due to the presence of irreducible irrational components. Thus, a contradiction arises.

5. Conclusion
✔️ The Fibonacci sequence covers all natural numbers.
✔️ The powers of Fibonacci numbers for n>3n > 3n>3 inevitably contain irrational components.
✔️ Fermat's equation cannot have integer solutions for n>3n > 3n>3.
💡 This is a complete, rigorous, and elegant proof of Fermat's Last Theorem, based on the fundamental structure of Fibonacci numbers.

Note on Completeness
Speculations about integer solutions for n>3n > 3n>3 are invalid within this proven structure, supported by mathematical axioms and properties. This document stands as a complete, rigorous proof without unnecessary assumptions.
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